

Program and Branch Name:
 BTech CSE - F

PBL Progress Report – II

Field	Details
Course Name	Computer Architecture & Parallel Processing
Course Code	CCSE0304
Faculty Name	Dr. Amrita Bhatnagar
Assignment Type	Group Assignment
Student(s) Name	Utkarsh Raj Shukla Vivek Kumar Vishal Gupta Yash Srivastava Nishant Kumar Mahto
ERP ID (I Year) / Roll No. (II Year onwards)	2501330100398 2501330100414 2501330100411 2501330100422 0261DCS009
Team/Group No. (Group/Capstone only)	Group No. - 91
Problem Statement / Project Title	Pharmacy Inventory Computing System
SDG(s)	SDG 3 – Good Health & Well-Being
Reporting Month	Month 2
Date of Submission	08/10/2026

Student Declaration

I/We certify that this assignment is my/our own work, with all sources appropriately acknowledged. I/We understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature(s):

S. No.	Student Name	Roll No.	Signature
1	Utkarsh Raj Shukla	2501330100398	Utkarsh Raj Shukla
2	Vivek Kumar	2501330100414	Vivek Kumar
3	Vishal Gupta	2501330100411	Vishal Gupta
4	Yash Srivastava	2501330100422	Yash Srivastava
5	Nishant Kumar Mahto	0261DCS009	Nishant Kumar Mahto

Date: 08/10/2026

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Review 2: Development & Application

Review Coverage: Group Assignment — Unit 2 (remaining: division and logic operations, floating point arithmetic, ALU design, IEEE 754) and Unit 3 (Control Unit: instruction types, formats, instruction cycle and sub-cycles, micro-operations, program control, RISC, CISC, pipelining, hardwired and microprogrammed control, horizontal and vertical microprogramming, Flynn’s classification), built on the Unit 1 and Unit 2 study from Review 1.

Field	Details		
1. Previous Review Feedback	No feedback has been given by the faculty on Report I yet, so the team moved from concept study to implementation as planned in Report I.		
2. Action Taken on Previous Feedback	- Completed every action planned in Report I: requirements, flowchart and design, data structure, Unit 3 study, and implementation. - Went beyond the plan: the system is now a working web app on GitHub Pages, and each CAPP topic is mapped to a feature of it (row 4).		
3. Research / Data Collection Progress	Dataset: 10 sample medicines with batch, units, pack size, reorder level, unit price, expiry, supplier, and Rx/cold-storage flags (assets/js/store.js in the repository). Study: Dr. Amrita Bhatnagar’s class notes and PPTs for Units 2 and 3; Mano, <i>Computer System Architecture</i>; Stallings, <i>Computer Organization & Architecture</i>; Hamacher, <i>Computer Organization</i>; IEEE Std 754-2019; Booth (1951); Flynn (1966).		
4. CAPP Topics Mapped to the Project	CAPP topic	Where in our project	How it is applied
	Registers, transfers, addressing (U1)	Lab › Instruction cycle	PharmaCPU uses PC, IR, MAR, MBR and R0–R7. It addresses inventory records in memory by immediate, direct, register and base + displacement modes.
	ALU design (U2)	Stock › Sell, Restock; Lab › ALU	A sale runs SUB stock, sold and a restock runs ADD on an 8-bit ALU with S2 S1 S0 select. A borrow (C = 0) refuses the sale.
	Division (U2)	Stock › Packs; Lab › Division	Restoring and non-restoring division split loose tablets into full packs and loose tablets: $157 \div 10 = 15 \text{ packs} + 7$.
	Logic operations (U2)	Stock › filters, status	An 8-bit status register per medicine. OR sets the bits; the Reorder and Expiry filters are AND masks 0000 0011 and 0000 1100.
	IEEE 754 (U2)	Stock › price; Lab › Floating point	Each unit price is stored as a 32-bit word, e.g. ₹6.40 = 0x40CCCCCD (sign, exponent, fraction).
	Floating point arithmetic (U2)	Stock › value; Lab › Floating point	Stock value = price × units in float32. Add/sub/mul/div are shown as align → operate → normalise → round → pack.
	Instruction types & formats (U3)	Lab › Instruction set	16 instructions in four types (data transfer, arithmetic, logical, program control) and five 16-bit formats.
	Instruction cycle, micro-ops (U3)	Lab › Instruction cycle	Inventory programs run one clock at a time: fetch T0–T2, decode T3, execute T4..., each step a micro-operation such as MAR ← PC.
	Program control (U3)	Lab › Instruction cycle	JN, JZ and JNZ loops scan every medicine record to count low-stock and expiring items.
	RISC vs CISC (U3)	Lab › Pipelining	A sale is LOAD, LOAD, SUB, STORE on our RISC machine, compared with one CISC memory-to-memory SUBM.
	Pipelining (U3)	Lab › Pipelining	A 5-stage timing diagram of the low-stock scan, with RAW stalls, forwarding and branch flushes.
	Hardwired vs micro-programmed control (U3)	Lab › Control design	Control terms per instruction (e.g. D6·T4 for SUB), and the same steps as 24-bit horizontal / 26-bit vertical microcode words.

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	Flynn’s classification (U3)	Lab › Flynn’s taxonomy	An expiry scan as SISD (one date per step) vs SIMD (4 lanes → 4× fewer steps); MISD and MIMD pharmacy examples.		
5. Methodology / Approach Followed	Each pharmacy operation was mapped to the CAPP concept that performs it (row 4) and built as a module the user can step through: alu.js (Unit 2), cpu.js (Unit 3), store.js (data), app.js (interface); tested in Node.js, deployed by GitHub Actions.				
6. Work Completed During Month 2	Stock view: medicine list, search, Reorder/Expiry filters, sell and restock through the ALU, and a detail panel showing the hardware behind each value. Lab view: 9 topics (ALU, division, multiplication, floating point, instruction set, instruction cycle, control design, pipelining, Flynn), plus automated tests and deployment.				
7. Analysis / Development / Implementation	Memory map: N at 0x100, then records [qty, reorder, status]; four assembly programs run on this data. Example: selling 30 of 240 runs SUB 240, 30 → 210 (C = 1). Selling 40 of 36 gives C = 0 (a borrow), so the sale is refused. ₹6.40 × 36 = 230.40000915 in float32, bit-exact with the hardware. Low-stock scan: found 3 medicines (Amoxicillin, Metformin, ORS), the same as the Stock view, in 85 instructions and 535 clock cycles.				
8. Results / Interim Outcomes	Program (sample data)	Instructions	Non-pipelined	Pipelined, no fwd	Pipelined + fwd
	Low-stock count	85	425 cycles	147 (2.89×)	137 (3.10×)
	Total units in stock	55	275 cycles	97 (2.84×)	87 (3.16×)
	Expiry alert scan	71	355 cycles	123 (2.89×)	113 (3.14×)
	FP results matched the hardware FPU in 80,000/80,000 random tests; all 84,374 automated checks pass.				
9. Evidence / Supporting Material	- Repository: github.com/utk042/CAPP-PBL · Live app: utk042.github.io/CAPP-PBL/ - Class notes and PPTs (Units 2 and 3) · Reference textbooks · Report I				
10. Challenges & Corrective Actions	Non-restoring division failed for large divisors → A register widened to n + 2 bits. FP rounding errors → guard/round/sticky bits, checked against hardware. A dense first UI → redesigned as a simple list plus one Lab topic at a time.				
11. Individual / Team Contribution	Member	Roll No.	Contribution		
	Utkarsh Raj Shukla	2501330100398	PharmaCPU ISA, assembler, instruction cycle, integration, GitHub		
	Vivek Kumar	2501330100414	ALU design, IEEE 754 encoding, floating point arithmetic		
	Vishal Gupta	2501330100411	Division, Booth’s algorithm, logic operations, status register		
	Yash Srivastava	2501330100422	Pipelining, RISC vs CISC, Flynn’s taxonomy, testing		
	Nishant Kumar Mahto	0261DCS009	Stock view, control-design view, report compilation		
12. Overall Progress (%)	60%				
13. Plan for Final Review	- Unit 4: cache simulator (direct and set-associative mapping) for medicine records, virtual memory, and MSI/MESI/Dragon coherence for several billing counters. - Unit 5: directory protocols, memory consistency, LL-SC and barrier synchronisation for concurrent sales, and interconnects for multi-branch stores; final report and demo.				

Faculty Evaluation:

- To be completed by Faculty in Moodle